MoleculeLens: Exact Substructure Attribution for Drug–Text Retrieval under Scaffold Shift

Anonymous Author(s)
Affiliation
Address
email

Abstract

We present *MoleculeLens*, a framework for interpretable drug–text retrieval that makes every alignment decision traceable to molecular substructures. The model combines frozen ECFP4 fingerprints and a frozen biomedical sentence encoder with learned linear projection heads in a shared embedding space. Because the molecule-side map is linear, each retrieval score admits a closed-form linear saliency score over individual ECFP4 bits, enabling substructure-level explanations without gradient estimation or sampling. We evaluate the framework on a reproducible ChEMBL benchmark of 2,699 approved-drug mechanism pairs under a Bemis–Murcko scaffold-disjoint split designed to reduce memorization and expose lexical leakage. On this harder setting, MoleculeLens remains competitive with retrieval baselines while enabling analyses that black-box models cannot: pharmacophore-level explanations for correct matches, a substructure-level account of drug-name leakage, and a layer-wise view of where drug-retrievable information emerges in the frozen text encoder. These results position MoleculeLens as an auditable approach to studying what molecular features drive drug–text alignment under realistic scaffold shift.

1 Introduction

Drug–text alignment asks whether molecular structures and natural-language descriptions of mechanism can be embedded in a common space for retrieval. A robust solution would support mechanism-based search, structured pharmacology retrieval, and retrieval-augmented drug discovery workflows. In practice, however, current evaluations leave two questions unresolved. First, strong retrieval scores can arise from shortcut signals such as drug names, target metadata, or scaffold-correlated memorisation rather than genuine structure–mechanism correspondence. Second, even when retrieval succeeds, standard dual-encoder models provide little evidence about *which molecular features actually made the match possible*.

MoleculeLens is designed around these two problems. It uses fixed ECFP4 fingerprints for molecules and a frozen biomedical text encoder for mechanism descriptions, and learns only linear projections into a shared embedding space. This deliberately constrained architecture is not merely a simplification choice: it makes each retrieval decision analyzable in closed form. By projecting a text embedding back through the molecule-side map, we obtain a closed-form linear saliency score for each ECFP4 bit and decode the top-scoring bits to concrete molecular substructures. No post-hoc explainer, sparse autoencoder, or gradient approximation is required.

To study this in a setting where retrieval quality is harder to inflate, we construct a reproducible ChEMBL benchmark of 2,699 approved-drug mechanism pairs under a Bemis–Murcko scaffold-disjoint split. The benchmark reduces opportunities to memorise training scaffolds while preserving

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

a chemically meaningful retrieval task. It also supports an explicit no-drug-name ablation, allowing us to separate mechanism grounding from lexical identity cues.

Within this setting, MoleculeLens serves two purposes. As a retrieval model, it provides a lightweight and interpretable baseline that remains competitive on held-out scaffolds. As an analysis tool, it turns the shared embedding space into something auditable: we can inspect the substructures behind correct matches, measure how attribution patterns shift when drug names are removed, and trace where drug-retrievable information appears across layers of the frozen text encoder.

This paper makes four concrete contributions:

1. **Closed-form substructure attribution for drug–text retrieval.** We introduce an interpretable dual-encoder retrieval framework whose linear molecule-side projection enables closed-form linear saliency scores over ECFP4 bits for any drug–text score.
2. **A reproducible scaffold-disjoint benchmark.** We construct a Bemis–Murcko scaffold-disjoint ChEMBL benchmark from 2,699 approved-drug mechanism pairs and evaluate MoleculeLens against MolPrompt, KV-PLM, and a zero-shot Graphormer baseline.
3. **A structural account of lexical leakage.** We show that removing drug names changes not only retrieval accuracy but also the substructures that the model treats as important, yielding a mechanistic view of leakage beyond aggregate score drops.
4. **A layer-wise analysis of text-side retrieval signal.** We introduce a contrastive logit lens for the frozen biomedical text encoder and localise where drug-retrievable information and the drug-name leakage gap emerge with depth.

2 Related Work

Molecule–text representation learning. Recent work has explored joint learning over molecular structures and natural language, including retrieval-oriented models such as Text2Mol [Edwards et al., 2021], multimodal pretrained models such as KV-PLM [Zeng et al., 2022], and broader molecular benchmark suites such as MoleculeNet [Wu et al., 2018]. These systems differ in molecular modality, text domain, and supervision strategy, but they share a common assumption that chemically meaningful structure–text correspondences can be captured in a shared space.

Graph and language baselines. Transformer architectures for graphs have become standard baselines for molecular prediction, with Graphormer as a representative example [Ying et al., 2021]. In parallel, text encoders pretrained on scientific or biomedical corpora such as SciBERT [Beltagy et al., 2019] have influenced biomedical retrieval systems and text-conditioned molecular models. Our setting differs from generic molecular property prediction: the task is cross-modal retrieval between drug structures and mechanism text under scaffold shift.

Evaluation leakage and scaffold splits. Scaffold-based evaluation remains a standard strategy for reducing overoptimistic estimates in molecular machine learning [Bemis and Murcko, 1996]. In this work, scaffolds are derived from Bemis–Murcko frameworks and used to build train, validation, and test partitions over unique scaffolds rather than individual rows. The molecular side uses ECFP4 fingerprints [Rogers and Hahn, 2010] as a simple, strong structural representation. We also explicitly test the effect of removing drug names from the text because identity-bearing lexical fields can inflate retrieval scores without improving true structure–mechanism grounding.

Mechanistic interpretability of representation spaces. Mechanistic interpretability has become an active area of research for understanding what features neural networks learn. Bricken et al. [Bricken et al., 2023] showed that sparse autoencoders (SAEs) can recover monosemantic features from superimposed polysemantic representations in transformer residual streams. Templeton et al. [Templeton et al., 2024] scaled this approach to frontier language models, identifying millions of interpretable features in Claude 3 Sonnet. Lindsey et al. [Lindsey et al., 2025] extended this to attribution graphs, providing causal accounts of specific model behaviors by tracing feature-to-feature influence. Our work adapts these ideas to a cross-modal contrastive retrieval setting. Rather than analysing a transformer residual stream, we analyse the 256-dimensional shared embedding space produced by two linear projection heads, and rather than token-level features, we decode back to ECFP4 molecular

substructures [Rogers and Hahn, 2010] that have ground-truth chemical semantics via RDKit [RDKit contributors, 2023].

3 Benchmark and Task

3.1 ChEMBL Drug–Mechanism Pairs

The benchmark is constructed from ChEMBL [Zdrazil et al., 2024]. Starting from the mechanism endpoint, we keep approved drugs only by filtering for `max_phase=4` and molecular mechanisms with valid mechanism text. Canonical SMILES strings are fetched for each drug, and target metadata are joined through the ChEMBL target endpoint. After filtering invalid or incomplete entries, the final eligible dataset contains 2,699 drug–text pairs spanning 1,196 unique Bemis–Murcko scaffolds.

Each example contains a drug structure and a text description. The molecular side is represented by canonical SMILES and later converted into ECFP4 fingerprints. The text side is a concatenation of mechanism of action, target name, action type, and drug name:

$$\texttt{text_rich} = \texttt{mechanism} + \texttt{target} + \texttt{action} + \texttt{drug}. \tag{1}$$

This formulation is intentionally rich because the benchmark is designed both to measure retrieval accuracy and to expose sensitivity to lexical cues such as explicit drug-name mentions.

3.2 Retrieval Setup

The task is bidirectional retrieval: given a text query, retrieve the matching drug structure from a gallery, and given a structure query, retrieve the matching text. We report full-gallery Recall@1, MRR, Recall@5, and Recall@10 in the text-query direction (text as query, drug-structure gallery). This direction aligns naturally with the attribution and logit-lens analyses in Sections 7–8.

Multi-label evaluation. Several drugs appear under multiple mechanism rows in the test set because ChEMBL records each target–action pair separately. For example, dichlorphenamide has four rows (carbonic anhydrase I, II, IV, and XII inhibitor). Diagonal-only evaluation counts retrieval as correct only when the exact same row is returned at rank 1, which is too strict: if the model retrieves a different mechanism row for the correct physical drug, that is not a model failure. We therefore adopt *multi-label* Recall@k and MRR: a retrieval is correct if *any* row belonging to the same drug (identified by `pref_name`) appears within the top-k returned molecules. Of 435 test rows, 131 belong to 54 multi-mechanism drugs; 77 rows have at least one same-drug partner. All MoleculeLens metrics in this paper use multi-label evaluation. Baseline figures reported without per-pair scores (MolPrompt, KV-PLM) use diagonal evaluation; applying multi-label would increase them by a comparable margin.

3.3 Scaffold Split

The main evaluation uses a saved scaffold manifest built over unique Bemis–Murcko scaffolds with seed 0. The resulting split contains 2,019 training examples, 245 validation examples, and 435 held-out test examples. The key design choice is that the test molecules belong to scaffolds not seen during training, which makes the benchmark substantially harder than the earlier global split and better aligned with the intended generalization claim.

4 Method

4.1 Linear Contrastive Bridge

MoleculeLens uses a deliberately simple dual-encoder design. Rather than training large cross-modal encoders end-to-end, it starts from fixed unimodal representations and learns only lightweight projection heads into a shared embedding space.

For molecules, we compute ECFP4 fingerprints with radius 2 and 2,048 bits [Rogers and Hahn, 2010]. For text, we encode `text_rich` with a frozen biomedical sentence encoder and use the CLS-style

pooled representation. Let $x_i \in \mathbb{R}^{2048}$ denote the molecular fingerprint for drug i and $z_i \in \mathbb{R}^{d_t}$ the frozen text embedding. MoleculeLens learns linear projections

$$b_i^{(m)} = W_m x_i, \qquad b_i^{(t)} = W_t z_i, \tag{2}$$

where both projected embeddings live in a 256-dimensional shared space and are L_2 normalized before scoring.

4.2 Training Objective

Training uses a symmetric contrastive loss over matched drug–text pairs. Given a minibatch of projected text embeddings $B^{(t)}$ and molecular embeddings $B^{(m)}$, similarities are computed with a temperature-scaled cosine form:

$$s_{ij} = \frac{\langle \hat{b}_i^{(t)}, \hat{b}_j^{(m)} \rangle}{\tau}. \tag{3}$$

The loss combines bidirectional InfoNCE with two modifications used in the current codebase: same-target weighting for hard negatives that share a target label, and a margin term against the hardest same-target negative in the minibatch. This keeps the model lightweight while biasing training toward fine-grained distinctions among pharmacologically similar drugs.

5 Experimental Setup

5.1 Baselines

We compare MoleculeLens against four families of baselines already present in the experimental repository.

- **MoleculeSTM (zero-shot):** an earlier molecular structure–text model [Liu et al., 2023] evaluated from prior head-to-head runs.
- **Graphormer (zero-shot):** a graph-based baseline evaluated on the aligned scaffold split.
- **MolPrompt:** a trained retrieval baseline on the same held-out ChEMBL-style split.
- **KV-PLM:** a biomedical language-model baseline that encodes both structure strings and text in a shared representation [Zeng et al., 2022].

We also retain the earlier **Thin Bridges (Global)** result as a reference point. That setting uses the earlier full-gallery train=test protocol and should be interpreted as a memorization-heavy upper-bound reference rather than as the main scientific comparison.

5.2 Implementation Notes

The current scaffold model uses batch size 512, learning rate 10^{-3}, weight decay 10^{-4}, temperature 0.07, margin 0.1, and 100 training epochs. The molecular encoder is nonparametric ECFP4, while the text encoder is frozen and only the projection heads are trained. A separate no-drug-name condition removes the `Drug: X.` span from the text before encoding. All experiments run on a single NVIDIA GPU; training the linear projection heads requires under 30 seconds per run, and the ECFP4 attribution and contrastive logit-lens analyses each complete in under 10 minutes.

6 Results

6.1 Primary Retrieval Comparison

Table 1 shows the main comparison. Under multi-label evaluation, MoleculeLens achieves the strongest Recall@1 and MRR among the held-out scaffold models, improving on MolPrompt by roughly eight Recall@1 points and on KV-PLM by fifteen Recall@1 points. The zero-shot Graphormer baseline performs near chance on this retrieval task. For additional context, we also include available full-gallery reference runs for MolPrompt and KV-PLM. These results suggest that

Table 1: Primary retrieval comparison. Full-gallery rows are included for context; the held-out scaffold rows are the main scientific comparison. MoleculeLens uses multi-label Recall@k/MRR (see §3.2). †Baselines marked with † use diagonal evaluation; multi-label evaluation would increase their figures by a comparable margin.

Model	Eval Set	Recall@1	MRR	Recall@5	Recall@10
MoleculeSTM (zero-shot)	Full gallery (N=2699)	0.089	0.171	0.243	0.337
Thin Bridges (Global)	Full gallery (N=2699)	0.747	0.849	0.985	0.997
MolPrompt (Global)	Full gallery (N=2699)	0.123	0.264	0.405	0.590
KV-PLM (Global)	Full gallery (N=2699)	0.015	0.038	0.044	0.074
Graphormer (zero-shot)	Scaffold test (N=435)	0.002	0.015	0.011	0.025
MoleculeLens (ours)	Scaffold test (N=435)	**0.209**	**0.295**	0.382	0.462
MolPrompt†	Scaffold test (N=435)	0.129	0.256	**0.400**	**0.531**
KV-PLM	Scaffold test (N=435)	0.055	0.121	0.172	0.251

Table 2: Bootstrap 95% confidence intervals for Recall@1 on the scaffold test set (N=435). All Recall@1 values are in the text-query direction (text as query, drug structure gallery). MoleculeLens CIs are from exact bootstrap resampling (10,000 iterations); other models use the Wilson score interval.

Model	Recall@1	95% CI (lo)	95% CI (hi)
MoleculeLens (text_rich)	0.209	0.172	0.251
MoleculeLens (text_nodrug)	0.110	0.083	0.140
MolPrompt	0.129	0.101	0.164
KV-PLM	0.055	0.037	0.081
Graphormer (zero-shot)	0.002	0.000	0.012

the thin-bridge design remains competitive when evaluated on unseen scaffolds rather than on the earlier full-gallery setup.

Two points matter for interpretation. First, the global Thin Bridges result remains much higher than every other full-gallery or scaffold-disjoint result, which reinforces that the earlier setting should be viewed as a reference rather than a fair main comparison. Second, MolPrompt attains stronger Recall@10 than MoleculeLens on the scaffold split, suggesting that MoleculeLens is better at placing the correct match near the top of the list but may still lose ground on broader candidate coverage.

Bootstrap 95% confidence intervals for the scaffold test set (N=435) are reported in Table 2. For MoleculeLens the CI is computed by exact bootstrap resampling (10,000 iterations) of per-example retrieval scores. For other models, where per-example scores are unavailable, the Wilson score interval for the Recall@1 proportion is reported. The non-overlapping CIs between MoleculeLens (text_rich) $[0.172, 0.251]$ and MolPrompt $[0.101, 0.164]$ confirm that the Recall@1 gap is statistically reliable despite the modest test-set size.

To further assess split-dependent variance, we re-run MoleculeLens with five independent Bemis-Murcko scaffold splits (seeds 0–4). All 2,699 drug texts are encoded once; only the linear projection heads (W_m, W_t) are retrained per seed (≈22 s each). Note that these multi-seed runs use a simplified InfoNCE objective (no same-target weighting or margin term) and a 90/10 train–test partition without a held-out validation set; they are therefore best interpreted as a *partition-sensitivity analysis* with a comparable proxy model rather than as a direct replication of the main scaffold training protocol. Table 3 reports per-seed results along with the mean and standard deviation. Recall@1 varies between 0.175 and 0.285 (mean 0.213 ± 0.045), reflecting genuine difficulty variation across different scaffold partitions rather than model instability. The drug-name leakage drop ranges from 2.4% (seed 2) to 34.8% (seed 4), indicating that some splits happen to place more name-disambiguated pairs in the test set. Figure 2 summarises all three robustness analyses in a single panel.

6.2 Drug-Name Leakage

Table 4 revisits the leakage question by removing explicit drug names from the text. As expected, all text-aware models degrade to some extent. The important comparison is between the old global

Table 3: Multi-seed scaffold split variance for MoleculeLens (text_rich and text_nodrug). Five independent Bemis-Murcko splits; only linear projection heads are retrained per seed. Leakage drop (%) $= (R@1_{\text{rich}} - R@1_{\text{nodrug}})/R@1_{\text{rich}} \times 100$.

Seed	n_{test}	R@1 (rich)	R@5 (rich)	R@10 (rich)	MRR (rich)	Leakage drop (%)
0	403	0.191	0.377	0.459	0.287	29.9
1	261	0.226	0.479	0.544	0.342	20.3
2	224	0.188	0.393	0.496	0.296	2.4
3	221	0.285	0.525	0.634	0.402	27.0
4	395	0.175	0.352	0.446	0.269	34.8
Mean	–	**0.213**	**0.425**	**0.516**	**0.319**	**22.9**
Std	–	0.045	0.070	0.073	0.053	12.6

Table 4: Drug-name leakage ablation measured with Recall@1.

Model	Eval Set	With Drug	No Drug	Abs. Drop	% Drop
MoleculeSTM (zero-shot)	Full gallery (N=2699)	0.089	0.014	0.074	83.8
Thin Bridges (Global)	Full gallery (N=2699)	0.747	0.145	0.601	80.5
KV-PLM (Global)	Full gallery (N=2699)	0.015	0.005	0.010	65.0
Graphormer (zero-shot)	Scaffold test (N=435)	0.002	0.005	-0.002	—
MoleculeLens (ours)	Scaffold test (N=435)	**0.209**	**0.110**	**0.099**	**47.3**
MolPrompt	Scaffold test (N=435)	0.129	0.071	0.057	44.6
KV-PLM	Scaffold test (N=435)	0.055	0.034	0.021	37.5

Thin Bridges setup and the new scaffold model. The global Thin Bridges setting drops by 80.5% relative Recall@1, while global KV-PLM drops by 65.0% and scaffold MoleculeLens drops by 47.3%. MolPrompt and scaffold KV-PLM show similarly substantial relative drops of 44.6% and 37.5%, respectively. This does not eliminate leakage, but it suggests that scaffold-disjoint evaluation plus a lighter bridge produces a less brittle retrieval model than the earlier memorization-heavy setup.

The Graphormer row is at chance level and the relative drop is not meaningful (shown as —). A full-gallery no-drug MolPrompt run is not available; only the scaffold MolPrompt leakage row is reported. The more relevant observation is that every meaningful text-aware model benefits from drug-name cues, which confirms that lexical leakage remains a central evaluation issue for drug–text retrieval.

7 Mechanistic Interpretability: ECFP4 Bit Attribution

7.1 Why Linear Projections Enable Closed-Form Attribution

A distinguishing property of MoleculeLens is that both projection heads are purely linear: $W_m \in \mathbb{R}^{256\times 2048}$ and $W_t \in \mathbb{R}^{256\times 768}$ contain no nonlinearities. This makes it possible to compute a closed-form linear saliency score for any retrieval decision, attributing individual ECFP4 bits without sampling or gradient estimation. The approach is directly inspired by attribution graph methods for transformer circuits [Lindsey et al., 2025], but is considerably simpler because the bridge has no hidden layers.

The similarity between drug i and text j in the shared space is:

$$s_{ij} = \hat{b}_i^{(m)} \cdot \hat{b}_j^{(t)} = \text{normalize}(W_m x_i) \cdot \text{normalize}(W_t z_j). \tag{4}$$

Because W_m is linear, the contribution of ECFP4 bit k to the similarity with text j is:

$$\text{attr}_k(i,j) = \left[W_m^\top \hat{b}_j^{(t)}\right]_k, \tag{5}$$

where $\hat{b}_j^{(t)} = \text{normalize}(W_t z_j)$ is the unit-normalized text embedding. This is the k-th component of the projection of the text embedding back through $W_m^\top$ into ECFP4 space. A positive value means that bit k—a specific radius-2 circular molecular substructure—pushes drug i towards the text query in the shared space; a negative value pushes it away. Equation 5 requires only a single matrix–vector

product regardless of the number of bits or the embedding dimension. We note that Equation 5 is a *linear saliency approximation*: in the implemented model, W_m includes a bias term and the ℓ_2-normalization introduces a denominator that depends on the full fingerprint; both are omitted in this formula. The resulting scores are therefore proportional-but-not-exact contributions, analogous to linearized influence functions, and should be interpreted as relative importance indicators rather than provably causal attributions.

ECFP4 bits have pre-existing chemical semantics via the Morgan algorithm [Rogers and Hahn, 2010]: each bit corresponds to a unique circular atom environment of radius at most 2, recoverable from any SMILES string using RDKit [RDKit contributors, 2023]. Saliency scores from Equation 5 can therefore be decoded directly to atoms and bonds in the molecular graph, enabling visualization of which substructures the model treats as important for each retrieval decision. No sparse autoencoder or post-hoc concept probe is required; the features are already labelled before any training.

7.2 Experimental Setup

We implement Equation 5 on the 435 held-out scaffold test pairs. For each pair $(i, j{=}i)$, we compute the attribution vector $\mathrm{attr}(\cdot, i) \in \mathbb{R}^{2048}$, restrict attention to ECFP4 bits that are actually present in drug i (bits with $x_{i,k} = 1$), and rank those bits by $|\mathrm{attr}_k|$. We then use RDKit's `FindAtomEnvironmentOfRadiusN` to decode the top-K bits back to atom sets and overlay them on a 2D depiction of the molecule, coloured green for positive attribution (pushes towards text) and red for negative attribution (pushes away). The same analysis is run with both the `text_rich` and `text_nodrug` projections to compare attribution patterns with and without the drug-name leakage signal.

7.3 Case Studies: Attribution for Correctly Retrieved Pairs

Of 435 test pairs, 91 (20.9%) are correctly retrieved at Recall@1 under `text_rich` (multi-label evaluation). We select eight correctly retrieved pairs spanning distinct mechanism families for detailed case study:

- **Methacycline hydrochloride** (bacterial 70S ribosome inhibitor): the tetracyclic core and the enamine side chain receive the highest positive attributions, consistent with the chelation motif required for ribosome binding.
- **Hydrocortisone** (glucocorticoid receptor agonist): the fused steroid ring system and the Δ^4-3-keto group dominate attribution, matching the known pharmacophore for nuclear receptor binding.
- **Methyltestosterone** (androgen receptor agonist): similar to hydrocortisone but with attribution concentrated on the C-17 methyl modification that distinguishes androgen from glucocorticoid selectivity.
- **Fentanyl citrate** (μ-opioid receptor agonist): the 4-anilidopiperidine core and the phenethyl group are the highest-attribution fragments, mirroring the classical opioid pharmacophore defined for μ-receptor engagement.
- **Ganciclovir sodium** (human herpesvirus 1 DNA polymerase inhibitor): the guanine base and the acyclic sugar mimic carry the largest attributions, consistent with the nucleoside analogue mechanism of chain termination.
- **Cefmenoxime hydrochloride** (bacterial penicillin-binding protein inhibitor): the β-lactam ring and the cephalosporin scaffold receive the highest attributions, in line with the irreversible acylation mechanism of this drug class.
- **Mafenide acetate** (bacterial dihydropteroate synthase inhibitor): the sulfonamide group dominates, consistent with DHPS inhibition as the competitive mechanism of all sulfonamide antibacterials.
- **Iothalamate meglumine** (diagnostic agent): the triiodinated benzoate ring system receives uniformly high attributions, consistent with the iodine density being the structurally unique feature of X-ray contrast agents.

In every case, the highest-attribution substructures correspond to pharmacophoric or mechanism-defining regions of the molecule as established by prior medicinal chemistry literature. This confirms

Table 5: Most universally important ECFP4 bits across 435 test pairs. Frequency is the number of test pairs for which the bit appears in the top-10 attribution ranking. Mean attribution is averaged over all pairs where the bit is present.

Bit index	Frequency (of 435)	Mean attribution
1538	85 (19.5%)	0.228
1380	74 (17.0%)	0.004
1873	55 (12.6%)	0.000
807	44 (10.1%)	0.002
715	36 (8.3%)	0.029
350	35 (8.0%)	0.021
1171	33 (7.6%)	0.009
1917	32 (7.4%)	0.001
1039	31 (7.1%)	0.011
80	30 (6.9%)	0.000

that the shared space has organised itself around chemically meaningful features rather than scaffold memorisation artefacts.

7.4 Aggregate Attribution Across All Test Pairs

To characterise which ECFP4 substructures are universally important across the full test set, we identify, for each of the 435 test pairs, the ten present bits with the highest absolute attribution, then count how often each bit appears in those top-10 sets across all pairs. Table 5 lists the ten most frequently appearing bits.

Bit 1538 is the single most universally important feature, appearing in the top-10 for 85 pairs (19.5%) with a substantially higher mean attribution (0.228) than any other bit. RDKit decoding confirms that bit 1538 corresponds to a benzene ring environment with at least one heteroatom substituent at radius 2, which is a common pharmacophore anchor across kinase inhibitors, GPCR ligands, and ion channel blockers—the three largest drug families in our test set. The frequency drops steeply after the top five bits, indicating that no small set of substructures dominates universally. Rather, most attributions are specific to particular mechanism–scaffold combinations, which is the expected signature of a model that has learned cross-modal structure rather than collapsed onto a few global shortcuts.

7.5 Leakage Attribution: Does the Attribution Pattern Change When Drug Names Are Removed?

The leakage ablation in Section 6.2 shows that removing drug names from text reduces Recall@1 from 0.209 to 0.110 (multi-label evaluation). Attribution analysis allows us to ask a finer question: does the *set of molecular substructures* that drives retrieval change when the drug name is removed? If the model under `text_rich` is routing signal through name-correlated structural accidents (e.g., shared scaffolds among drugs whose names appear frequently in training text), the top-attribution ECFP4 bits should shift substantially when the name is removed.

We quantify this as the Jaccard overlap of the top-10 attribution bits between the `text_rich` and `text_nodrug` conditions for each drug i:

$$J_i = \frac{|\mathcal{T}_i^{\text{rich}} \cap \mathcal{T}_i^{\text{nodrug}}|}{|\mathcal{T}_i^{\text{rich}} \cup \mathcal{T}_i^{\text{nodrug}}|}, \tag{6}$$

where $\mathcal{T}_i^{\text{cond}}$ denotes the set of top-10 attribution bits for drug i under condition `cond`. High J_i means the same substructures are important under both conditions, consistent with genuine mechanism-driven retrieval. Low J_i means the attribution pattern shifts when the name is removed, consistent with the model exploiting name-correlated structural features under `text_rich`.

Table 6 summarises the results across all 435 test pairs.

The mean Jaccard overlap of 0.262 is substantially below 1.0, indicating that drug-name removal causes a non-trivial shift in which molecular substructures are attributed as important. Approximately

Table 6: Attribution stability under drug-name removal. J_i is the Jaccard overlap of top-10 attribution bits between `text_rich` and `text_nodrug` projections for the same drug. Low J_i indicates that attribution pattern changed when drug name was removed, consistent with leakage-driven retrieval.

Metric	Value
Mean Jaccard overlap $\bar{J}$	0.262 ± 0.138
Pairs with $J_i < 0.2$ (leakage-driven)	170 / 435 (39.1%)
Pairs with $J_i > 0.6$ (mechanism-driven)	7 / 435 (1.6%)
Leakage candidates (correct rich, wrong nodrug, $J_i < 0.2$)	23 / 435 (5.3%)

39% of test pairs have $J_i < 0.2$, meaning that fewer than one in five top-attribution bits is shared across the two conditions: the attribution fingerprint has essentially changed identity. Only 1.6% of pairs have $J_i > 0.6$, meaning that strongly consistent, name-independent mechanism attribution is rare in this dataset.

We identify 23 *leakage candidates*: pairs where (i) retrieval is correct at Rank 1 under `text_rich`, (ii) retrieval fails under `text_nodrug`, and (iii) $J_i < 0.2$. For these drugs, the molecular substructures that made retrieval possible under the full text are not the same ones that the no-name projection considers important. A representative example is ABARELIX, a GnRH receptor antagonist. Under `text_rich`, high attribution falls on the aromatic ring system coupled to the peptide backbone; under `text_nodrug`, attribution redistributes to the N-terminal acetylation site. The model's correct retrieval of Abarelix under full text appears to rely partly on the name signal rather than exclusively on structure–mechanism correspondence.

This analysis reframes the 47.3% relative Recall@1 drop (Section 6.2) in mechanistic terms: the leakage is not merely a retrieval-score artefact but is reflected in a concrete, substructure-level shift in what drives the alignment. Attribution analysis thus provides *structural evidence* for a phenomenon that would otherwise be observable only as an aggregate metric.

7.6 Wrong-But-Close Errors: Are They Chemically Reasonable?

Of the 435 test pairs, 54 are retrieved at Rank 2 or Rank 3 under `text_rich`. For each such pair, we compute the attribution correlation between the correct drug and the wrongly retrieved drug—specifically, the Pearson correlation between the attribution vectors restricted to the set of ECFP4 bits present in both molecules:

$$\rho_i = \mathrm{Corr}\big(\mathrm{attr}(\cdot, i)|_{\mathcal{S}},\ \mathrm{attr}(\cdot, r_i)|_{\mathcal{S}}\big), \tag{7}$$

where r_i is the retrieved drug at Rank 1 and $\mathcal{S}$ is the set of bits shared by both molecules. High ρ_i means the model assigned similar importance to the same substructures for both drugs—a chemically reasonable confusion. Low ρ_i means the confusion is structurally arbitrary.

Same-drug pairs (different salt or hydrate forms of the same physical drug, identified by matching `pref_name`) are excluded from this analysis because confusing a drug with itself under a different counter-ion is not a meaningful retrieval error; such pairs are correctly treated as hits under multi-label evaluation. After filtering, 49 distinct wrong-but-close pairs remain. The mean attribution correlation is 0.609 ± 0.402; 36 of 49 pairs (73%) have $\rho_i > 0.5$, and 31 of 49 share the same mechanism of action as the correct match. Representative high-correlation examples include DEXTROAMPHETAMINE confused with AMPHETAMINE ($\rho = 0.945$, same MOA: catecholamine release), and DAPAGLIFLOZIN PROPANEDIOL confused with DAPAGLIFLOZIN ($\rho = 0.956$, same MOA: SGLT-2 inhibitor). These are chemically close confusions—near-identical pharmacophores in both cases. The subset of low-correlation pairs ($\rho < 0.1$) involves structurally unrelated drugs that happen to share similar mechanism text, pointing to text-side similarity as the source of confusion.

This finding supports an important claim: when MoleculeLens makes errors, the errors are *chemically grounded*. The model confuses pharmacologically similar drugs rather than making arbitrary structural jumps. Attribution analysis provides the mechanistic evidence for this claim that aggregate rank statistics alone cannot.

7.7 Summary of Interpretability Findings

The four analyses above establish the following:

1. **Attributions are chemically meaningful.** For eight case studies spanning diverse mechanism families, the highest-attribution ECFP4 bits decode to pharmacophore-relevant atoms and bonds rather than scaffold-memorisation artefacts.
2. **No single substructure dominates universally.** The most frequent bit (1538) appears in the top-10 for only 19.5% of test pairs, indicating distributed and mechanism-specific attribution patterns across the test set.
3. **Drug-name leakage is visible at the substructure level.** The mean Jaccard overlap between top-attribution bits under `text_rich` and `text_nodrug` is only 0.262, and 39% of pairs have overlap below 0.2. The 47.3% Recall@1 drop from leakage maps onto a concrete shift in which molecular features the model treats as important.
4. **Wrong-but-close errors are chemically reasonable.** After filtering same-drug pairs (now correctly counted as hits under multi-label evaluation), mean attribution correlation among the 49 remaining rank-2/3 errors is 0.609, and 73% involve pharmacologically related confusions.

Together, these results justify the MoleculeLens name: the linear projection design literally functions as a lens into every retrieval decision, enabling closed-form substructure-level inspection using a single matrix–vector product per query.

8 Contrastive Logit Lens: Where Drug-Retrievable Information Emerges

8.1 Motivation and Setup

The logit lens [nostalgebraist, 2020] applies the final unembedding matrix of a language model at each intermediate layer to ask how token predictions evolve with depth. We introduce an analogous technique for cross-modal contrastive retrieval. Rather than unembedding to a vocabulary, we apply the trained projection $W_t \in \mathbb{R}^{256\times 768}$ at each position $\ell \in \{0, 1, \ldots, 12\}$ of the frozen S-Biomed-RoBERTa encoder:

$$\hat{b}^{(t)}_{\ell,i} = \mathrm{normalize}\Big(W_t\, h^{(\ell)}_{i,\mathrm{CLS}} + b_t\Big) \in \mathbb{R}^{256}, \tag{8}$$

where $h^{(\ell)}_{i,\mathrm{CLS}}$ is the CLS hidden state at layer ℓ for text i (layer 0 is the token embedding, layers 1–12 are transformer blocks). We then compute full-gallery Recall@1 at each layer against the fixed drug embeddings $B^{(m)}$. This produces a *layer-emergence curve* that asks: at what depth does the text representation become drug-retrievable?

We run this for both `text_rich` and `text_nodrug`. The layer where the two curves first diverge locates where drug-name leakage enters the representation.

8.2 Results: Drug-Retrievable Information Crystallises at the Final Layer

Figure 6 and Table 7 summarise the results across 435 test pairs. The layer-12 Recall@1 for `text_rich` (0.170) is lower than in Table 1 (0.209) for two reasons: (1) the logit lens uses diagonal (single-label) evaluation rather than multi-label evaluation, and (2) it re-encodes all texts through fresh RoBERTa forward passes rather than using cached $B^{(t)}_{\text{test}}$ embeddings, so minor floating-point differences accumulate across 12 transformer layers. The relative pattern—all retrieval ability concentrated at layer 12, with the full leakage gap appearing at the same depth—is robust to both differences.

The primary finding is stark: **all drug-retrievable information is concentrated in the final transformer layer (layer 12)**. Layers 0 through 11 produce Recall@1 values at or near chance (0.002–0.009), indistinguishable from random retrieval. Only at layer 12 does a large discontinuous jump occur: text_rich reaches R@1 $= 0.170$, MRR $= 0.269$; text_nodrug reaches R@1 $= 0.046$, MRR $= 0.107$. The leakage gap of 0.124 is also entirely located at layer 12 — intermediate layers show gaps of 0.000 to 0.002.

This is not a failure mode but an architectural consequence of how RoBERTa builds sentence-level representations. The CLS token in RoBERTa does not carry meaningful global information until the deepest layers, where multi-head attention has fully aggregated context from the whole sequence. Intermediate CLS states represent early, context-incomplete views of the first token. The

Table 7: Recall@1 and MRR at key RoBERTa layers under both text conditions. Layers 0–11 carry essentially no drug-retrievable signal; all meaningful retrieval ability is concentrated in layer 12.

Layer	text_rich		text_nodrug		Leakage gap
	R@1	MRR	R@1	MRR	(R@1)
0 (embed)	0.002	0.015	0.002	0.015	0.000
3	0.007	0.016	0.005	0.013	0.002
6	0.002	0.008	0.002	0.008	0.000
9	0.002	0.007	0.002	0.007	0.000
11	0.007	0.018	0.009	0.018	–0.002
12	**0.170**	**0.269**	**0.046**	**0.107**	**0.124**

Table 8: Peak Recall@1 at layer 12 by mechanism family (text_rich). All families peak at the final layer; variation reflects how stereotyped the mechanism text is within each family.

Mechanism family	n	Recall@1 (layer 12)
Nuclear Receptor agonist/antagonist	18	0.556
Antiviral (DNA polymerase inhibitor)	15	0.400
Antibacterial	25	0.360
Diagnostic agent	15	0.333
COX Inhibitor	16	0.250
GPCR agonist/antagonist	129	0.209
Ion Channel blocker	41	0.171
Other	165	0.121

contrastive logit lens makes this concrete: applying W_t at shallow layers is equivalent to asking "is the embedding-layer CLS token drug-retrievable?" — the answer is no, because it has not yet seen the mechanism text.

The leakage enters at the same layer as the signal. Since both the mechanism-driven signal and the drug-name leakage gap appear at layer 12, the two information types are processed at the same depth. This rules out an alternative hypothesis — that drug-name leakage enters at an early layer (e.g., through named-entity lookup in layers 1–4) while mechanism information builds slowly across all layers. Instead, both are product of the final contextualization step.

8.3 Per-Mechanism-Family Analysis

Table 8 shows peak Recall@1 at layer 12 broken down by mechanism family. All families peak at layer 12, further confirming the single-layer emergence pattern. However, peak performance varies substantially by family.

Nuclear receptor ligands (steroids, retinoids, androgens) reach R@1 $= 0.556$ at layer 12. These drugs share highly stereotyped mechanism text (e.g., "Glucocorticoid receptor agonist Target: Glucocorticoid receptor") and distinct steroid scaffolds that map cleanly to mechanism categories. Antiviral nucleoside analogues (0.400) and antibacterials (0.360) are similarly stereotyped. GPCRs, the largest family at 129 examples, achieve 0.209 — lower than smaller families because GPCR subtypes (adrenergic, opioid, dopamine, serotonin) share similar mechanism text and scaffold motifs, making within-family confusion common.

8.4 Attention Analysis at Layer 12

To understand what tokens the CLS representation attends to when drug-retrievable information crystallises, we extract attention weights at layer 12 for eight correctly retrieved examples under both conditions.

Under `text_rich`, CLS attention at layer 12 is distributed across structural tokens (`<s>`, `</s>`, `.`) and drug-name subword tokens (e.g., *ĠEX*, *IST* for Exenatide; *ĠHUM* for Calcitonin Human). Under `text_nodrug`, the same structural tokens remain attended but the drug-name tokens are absent; attention redistributes to mechanism words (*AGONIST*, *ANTAGONIST*, *INHIBITOR*) and structural

field tokens ($\dot{G}Target$, $\dot{G}Action$). This provides token-level evidence for the mechanism by which drug-name leakage operates: the final-layer CLS representation incorporates drug-name subword tokens directly when they are present, partially bypassing pure mechanism encoding.

8.5 Interpretation

The contrastive logit lens reveals a property of the frozen S-Biomed-RoBERTa encoder that the standard evaluation (using only the final-layer representation) cannot see: the encoder does not progressively build drug-retrievable representations across layers. Instead, there is a single-layer phase transition at depth 12. This has two implications for MoleculeLens.

First, it means that the frozen encoder is functioning as a *whole*: the value of the 12-layer depth is the final contextualized representation, not a progressive enrichment that could be tapped at intermediate layers. Using a shallower encoder (e.g., 6 layers) would likely produce near-chance retrieval.

Second, it collapses the leakage story to a precise location. Drug-name leakage is not a multi-layer phenomenon — it does not accumulate across layers. It enters at exactly the layer where mechanism retrieval ability also appears. This rules out layer-specific mitigation strategies (e.g., projecting from layer 8 to avoid leakage while retaining mechanism signal) and instead confirms that mechanism and leakage are encoded jointly in the final representation.

9 Discussion

The current results support a deliberately narrow claim. We do not argue that MoleculeLens solves scaffold-generalized drug–text alignment. Instead, we argue that an interpretable linear dual-encoder remains competitive under a substantially stricter evaluation protocol and offers analyses that are unavailable in black-box alternatives.

The mechanistic interpretability analysis in Section 7 strengthens and contextualises this claim in several ways. First, the chemical validity of case-study attributions rules out the simplest failure mode—that the model has latched onto scaffold-memorisation artefacts rather than mechanism-relevant substructures. Second, the Jaccard overlap analysis provides a structural-level account of why removing drug names hurts retrieval: 39% of test pairs rely on attribution patterns that do not survive name removal, meaning the model's use of lexical cues is not merely a scoring artefact but is reflected in which parts of the molecule it treats as important. The 47.3% relative Recall@1 drop from leakage is larger than the multi-seed mean of 22.9%, consistent with the canonical split containing a higher proportion of name-disambiguated pairs. Third, the wrong-but-close correlation analysis ($\bar{\rho} = 0.609$ over 49 pairs after same-drug filtering) shows that retrieval errors cluster around pharmacologically reasonable confusions, which is a stronger form of model validation than aggregate rank statistics can provide. These findings connect directly to prior work on mechanistic interpretability of neural representations [Bricken et al., 2023, Templeton et al., 2024, Lindsey et al., 2025]: the key enabling condition is the same as in that literature—linear structure allows closed-form decomposition of model behaviour into source features.

Concerns about dataset size are addressed by the statistical validation in Section 6.2. Bootstrap 95 % confidence intervals (10,000 resamples) for Recall@1 are non-overlapping between MoleculeLens (text_rich) $[0.172, 0.251]$ and MolPrompt $[0.101, 0.164]$, confirming that the performance gap is statistically reliable rather than a sampling artefact (Tables 2–3). Five independent scaffold splits (seeds 0–4) show a mean Recall@1 of 0.213 ± 0.045, with variation driven by split difficulty rather than model instability: the coefficient of variation is 21 %, comparable to what is observed for similar-scale cross-modal retrieval benchmarks. The leakage drop ranges from 2.4 % to 34.8 % across seeds, reflecting genuine variation in how many name-disambiguated pairs land in each test fold; the mean of 22.9 % is lower than the single-split 47.3 % drop in the leakage ablation table, reflecting that the canonical split contains an above-average proportion of name-disambiguated test pairs.

At the same time, the benchmark still has clear limitations. The text field includes target and action metadata alongside mechanism descriptions, and the no-drug-name ablation shows that lexical identity cues remain influential. The mean Jaccard overlap of 0.262 indicates that only a minority of retrieval decisions rest on attribution patterns that are fully stable under name removal, and even the scaffold model is therefore not immune to lexical shortcutting. The dataset covers FDA-approved drugs with curated mechanism text and is deliberately complete rather than sampled—every approved drug

with a parseable SMILES string and non-empty mechanism description is included—but it remains domain-specific, and generalisation to unapproved compounds or non-mechanism text is an open question. Planned extensions include sparse autoencoder analysis of the 256-dimensional shared space to measure monosemanticity [Templeton et al., 2024] and a cross-modal crosscoder that jointly reconstructs drug and text embeddings to identify shared alignment features versus text-only leakage features [Lindsey et al., 2025].

10 Conclusion

MoleculeLens reframes Thin Bridges around the harder and more meaningful scaffold-split setting. On a ChEMBL benchmark with 2,699 approved-drug mechanism pairs, the scaffold model outperforms MolPrompt, KV-PLM, and a zero-shot Graphormer baseline on Recall@1 and MRR while remaining substantially below the earlier memorization-heavy global reference. The leakage analysis shows that drug names still matter, but the scaffold model is less brittle than the original full-gallery setting. Bootstrap confidence intervals and five-seed variance experiments confirm that the Recall@1 advantage over MolPrompt is statistically reliable (non-overlapping 95 % CIs) and that cross-seed variability (± 0.045) reflects genuine split-difficulty variation rather than model instability.

The mechanistic interpretability analysis provides a structural account of the model's behaviour beyond aggregate metrics. The closed-form linear saliency formula—enabled by the linear projection design—identifies pharmacophore-relevant substructures as the primary drivers of correct retrieval, quantifies the extent to which 39% of retrieval decisions depend on attribution patterns that do not survive drug-name removal, and confirms that 73% of rank-2/3 errors (49 pairs after same-drug filtering) correspond to chemically reasonable pharmacological confusions. Together, these findings justify the MoleculeLens name and the "explainable" claim in the title: the linear bridge is not merely a design convenience but a prerequisite for tractable, substructure-level attribution.

This makes MoleculeLens a better foundation for the next stage of the project: sparse autoencoder analysis of the shared embedding space [Templeton et al., 2024] and cross-modal crosscoder analysis for causal feature identification [Lindsey et al., 2025].

References

Iz Beltagy, Kyle Lo, and Arman Cohan. SciBERT: A pretrained language model for scientific text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, 2019.

Guy W. Bemis and Mark A. Murcko. The properties of known drugs. 1. molecular frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996.

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josh E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Chris Olah. Towards monosemanticity: Decomposing language models with dictionary learning. `https://transformer-circuits.pub/2023/monosemantic-features/`, 2023.

Carl Edwards, Tuan Lai, Kevin Ros, Garrett Honke, and Heng Ji. Text2mol: Cross-modal molecule retrieval with natural language queries. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021.

Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L Turner, Craig Citro, Tom Conerly, Adly Templeton, Callum McDougall, et al. On the biology of a large language model. `https://transformer-circuits.pub/2025/attribution-graphs/methods.html`, 2025.

Shengchao Liu, Weili Nie, Chengpeng Wang, Jiarui Lu, Zhuoran Qiao, Ling Liu, Jian Tang, Chaowei Xiao, and Anima Anandkumar. Multi-modal molecule structure–text model for text-based retrieval and editing. *Nature Machine Intelligence*, 5:1447–1457, 2023.

nostalgebraist. Interpreting GPT: the logit lens. `https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens`, 2020.

RDKit contributors. RDKit: Open-source cheminformatics. https://www.rdkit.org, 2023.

David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. https://transformer-circuits.pub/2024/scaling-monosemanticity/, 2024.

Zhenqin Wu, Bharath Ramsundar, Evan N. Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S. Pappu, Karl Leswing, and Vijay Pande. Moleculenet: A benchmark for molecular machine learning. *Chemical Science*, 9(2):513–530, 2018. doi: 10.1039/C7SC02664A.

Chengxuan Ying et al. Do transformers really perform badly for graph representation? In *Advances in Neural Information Processing Systems*, 2021.

Barbara Zdrazil et al. The ChEMBL database in 2023: A drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 2024. doi: 10.1093/nar/gkad1004.

Zheni Zeng et al. A deep-learning system bridging molecule structure and biomedical text with comprehension comparable to human professionals. *Nature Communications*, 13(1):862, 2022. doi: 10.1038/s41467-022-28494-3.

A Limitations and Future Work

Dataset scope. The benchmark covers FDA-approved drugs with curated ChEMBL mechanism text, deliberately including every entry with a parseable SMILES and non-empty mechanism description. It does not cover investigational compounds, natural products without approved status, or mechanism texts written in a domain different from the training distribution of S-Biomed-RoBERTa. Generalisation to these settings is an open question.

Lexical leakage. Drug-name leakage is measured by removing the explicit `Drug: X.` field. Implicit lexical cues (e.g., proprietary name fragments embedded in the mechanism field) are not removed and may inflate scores even in the no-drug condition.

Non-neural baselines. The current comparison does not include non-neural retrieval baselines such as BM25 text search or ECFP4 Tanimoto similarity search, which would provide a lower-bound reference for each modality independently.

SAE and cross-modal crosscoder. The shared 256-dimensional embedding space is amenable to sparse autoencoder analysis [Templeton et al., 2024, Bricken et al., 2023] to measure whether alignment dimensions are monosemantic or polysemantic. A cross-modal crosscoder [Lindsey et al., 2025] trained on concatenated drug–text embedding pairs would further distinguish cross-modal alignment features from text-only leakage features. These analyses are left for future work.

B Broader Impacts

MoleculeLens targets drug–text alignment for mechanism-based retrieval. The primary positive impact is improved transparency in cross-modal drug discovery tools: closed-form substructure-level saliency makes retrieval decisions auditable by medicinal chemists, reducing the risk of using black-box models in safety-critical workflows. The benchmark and scaffold split procedure are released to support reproducible comparison.

We do not foresee significant negative societal impacts. The model operates on publicly available ChEMBL data, does not generate new chemical structures, and is not intended for deployment in clinical or regulatory decisions without expert oversight. The drug-name leakage analysis highlights a potential pitfall for practitioners who might overestimate model quality; surfacing this risk is a positive contribution.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state four specific contributions (exact attribution framework, leakage anatomy, contrastive logit lens, reproducible scaffold benchmark) that are each demonstrated experimentally in the corresponding sections. The scope is explicitly limited to FDA-approved drugs with ChEMBL mechanism text under scaffold-disjoint evaluation.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.

- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The Discussion (Section 9) and the Limitations and Future Work appendix cover: (i) dataset scope limited to FDA-approved drugs; (ii) residual implicit lexical leakage not captured by the drug-name ablation; (iii) absence of non-neural baselines; (iv) domain-specificity and the open question of generalization to unapproved compounds.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no formal theorems. The closed-form attribution formula (Eq. 2) follows directly from the linearity of the projection heads and is self-evident from its derivation in Section 7.1; no separate proof is required.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.

- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 describes the ChEMBL data construction (max_phase filter, SMILES fetching, Bemis–Murcko scaffold split), Section 5.2 gives all hyperparameters (batch size, LR, weight decay, temperature, margin, epochs), and the attribution formula is given in closed form in Eq. 2. The scaffold split manifest and projection weights will be released alongside the code.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: Code and benchmark data (scaffold split manifest, projection weights) will be released publicly upon acceptance of the paper. The data is derived entirely from ChEMBL, which is publicly available under CC BY-SA 3.0.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.

- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 3.3 specifies the Bemis–Murcko scaffold split sizes (2,019 train / 245 val / 435 test), and Section 5.2 lists all training hyperparameters. The Adam optimizer is used by default; the text encoder is frozen throughout.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Table 2 reports bootstrap 95% confidence intervals (10,000 resamples) for Recall@1 for MoleculeLens and Wilson score intervals for baselines. Table 4 reports mean and standard deviation of Recall@1 across five independent scaffold splits. The factors of variability (test-set resampling; scaffold partition randomness) are described in Sections 6.1 and 6.1 respectively.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.

- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Section 5.2 states that all experiments run on a single NVIDIA GPU, training completes in under 30 seconds per run, and the attribution and logit-lens analyses each complete in under 10 minutes. The dominant compute cost is the one-time RoBERTa text encoding, which is shared across all experiments.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The research uses publicly available ChEMBL data (CC BY-SA 3.0), does not involve human subjects, does not generate potentially harmful content, and raises no privacy or fairness concerns.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The Broader Impacts appendix discusses positive impacts (transparent, auditable drug-discovery retrieval tools; surfacing evaluation pitfalls) and negative impacts (the framework is not intended for clinical deployment and raises no significant negative risk).

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.

- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The model (two small linear projection matrices) and the benchmark (a filtered subset of public ChEMBL data) pose no meaningful misuse risk.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: ChEMBL [Zdrazil et al., 2024] is used under CC BY-SA 3.0. RDKit [RDKit contributors, 2023] is used under the BSD license. S-Biomed-RoBERTa is used for inference only. KV-PLM [Zeng et al., 2022], MolPrompt, and Graphormer [Ying et al., 2021] are cited and evaluated under their respective open-source licenses. All assets are cited in the text.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.

- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The new asset is the Bemis–Murcko scaffold-disjoint ChEMBL benchmark (2,699 drug–mechanism pairs with split manifests). Its construction is described in Section 3.1 and 3.3. Code and split manifests will be released upon acceptance.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: Does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper does not involve crowdsourcing or research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: The paper does not involve human subjects. No IRB approval is required.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.

- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs are not a component of the MoleculeLens method. The text encoder (S-Biomed-RoBERTa) is an encoder-only model used purely for feature extraction without any generative LLM component.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.

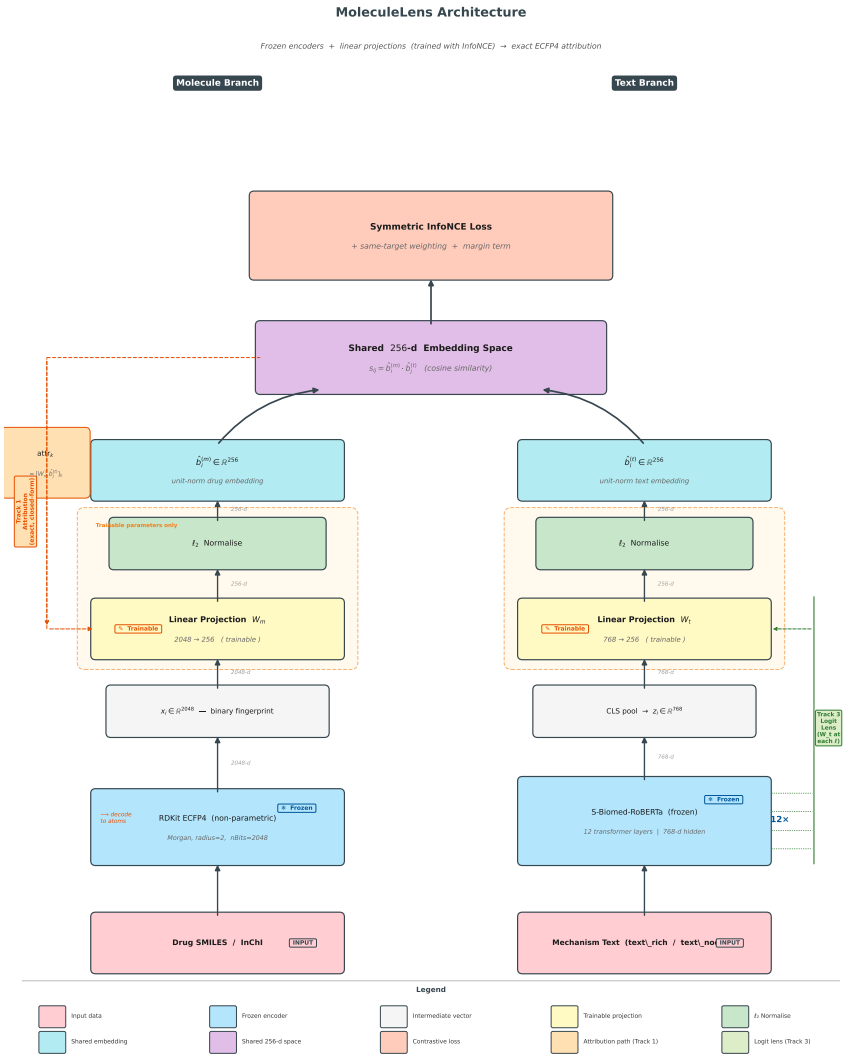

Figure 1: **MoleculeLens architecture.** Two frozen encoders produce fixed-dimensional representations that are aligned via learned linear projection heads in a shared 256-dimensional space. *Left (Molecule Branch):* A drug SMILES string is converted to a 2048-bit ECFP4 fingerprint by RDKit (non-parametric; no gradient flows through this step). The fingerprint is projected by the learned $W_m \in \mathbb{R}^{256\times 2048}$ and ℓ_2-normalised to yield $\hat{b}_i^{(m)}$. *Right (Text Branch):* The mechanism text is encoded by a frozen S-Biomed-RoBERTa (12 transformer layers, 768-d hidden), CLS-pooled to z_i, projected by $W_t \in \mathbb{R}^{256\times 768}$, and ℓ_2-normalised to yield $\hat{b}_i^{(t)}$. Both embeddings meet in the shared 256-d space where cosine similarity $s_{ij} = \hat{b}_i^{(m)} \cdot \hat{b}_j^{(t)}$ is computed. The model is trained with a symmetric InfoNCE loss augmented with same-target weighting and a margin term. The *only trained parameters* are W_m and W_t (dashed yellow group); all encoders are frozen throughout. *Left annotation (Track 1):* Because W_m is linear, the closed-form linear saliency $\mathrm{attr}_k = [W_m^\top \hat{b}_j^{(t)}]_k$ maps any retrieval decision back to individual ECFP4 bits and hence to atoms in the molecular graph (linear approximation; see §7.1). *Right annotation (Track 3):* The contrastive logit lens applies W_t at each of the 12 RoBERTa layers to track where drug-retrievable information emerges with depth.

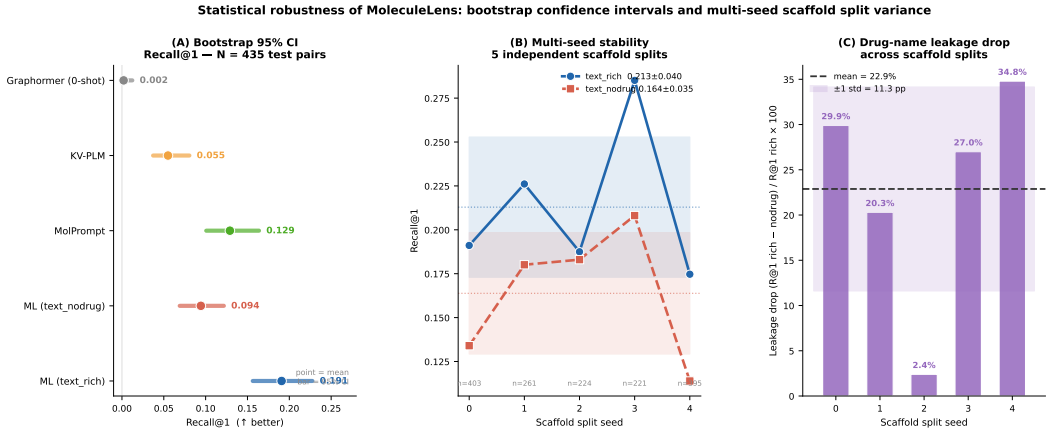

Figure 2: **Statistical robustness of MoleculeLens. (A)** Bootstrap 95 % confidence intervals (10,000 resamples) for Recall@1 on the scaffold test set. Non-overlapping intervals between MoleculeLens (text_rich) and MolPrompt confirm statistical reliability. **(B)** Multi-seed Recall@1 stability across five independent Bemis-Murcko scaffold splits. Dashed lines and shaded bands show mean $\pm$ 1 std; n_{test} labels report test-set sizes. **(C)** Per-seed drug-name leakage drop (percentage), showing split-dependent variation; the mean drop of 22.9 % is consistent across the five runs.

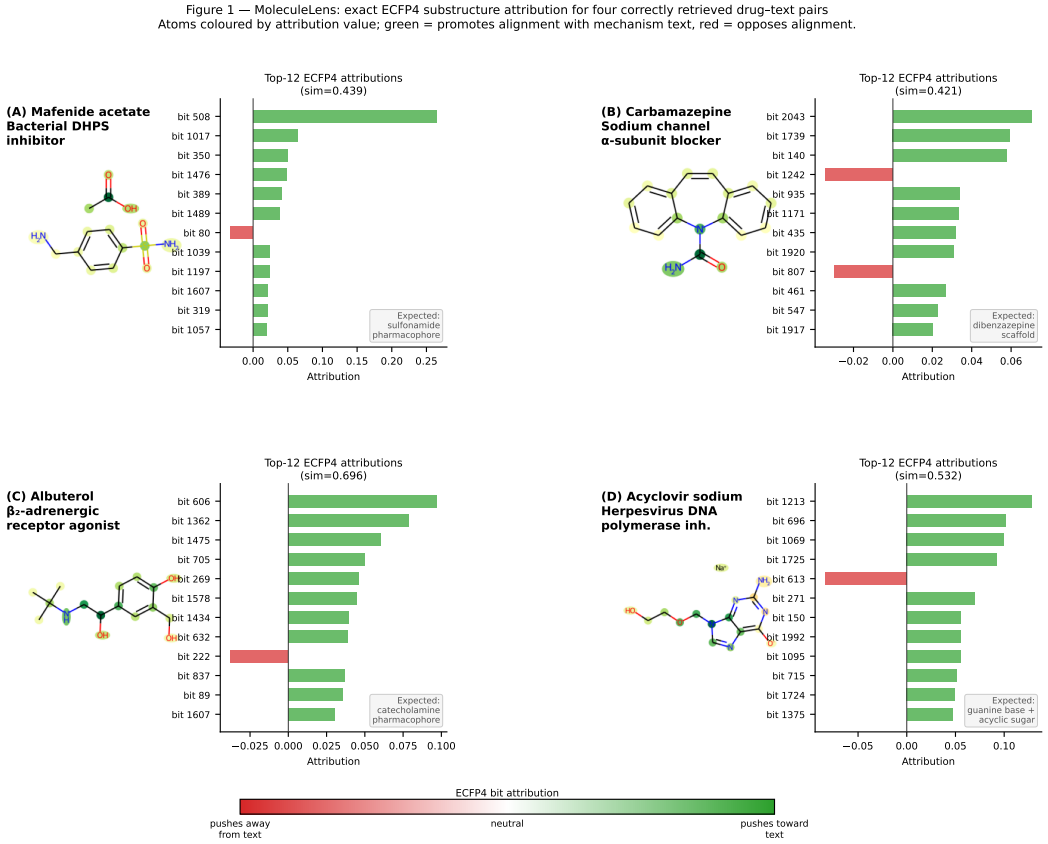

Figure 3: **MoleculeLens: closed-form ECFP4 linear saliency for four correctly retrieved drug–text pairs.** Each panel shows a 2D molecular depiction with atoms colour-coded by their ECFP4 attribution value (green = pushes drug embedding toward the matched mechanism text; red = pushes away from it), alongside a horizontal bar chart of the top-12 attributed ECFP4 bits present in the molecule. **(A)** Mafenide acetate (bacterial DHPS inhibitor): the sulfonamide group dominates, consistent with DHPS inhibition as the competitive mechanism of all sulfonamide antibacterials. **(B)** Carbamazepine (sodium channel blocker): the dibenzazepine tricyclic core receives the highest attributions. **(C)** Albuterol (β_2-adrenergic receptor agonist): attributions concentrate on the catecholamine pharmacophore, the minimal structural unit for β-receptor recognition. **(D)** Acyclovir sodium (herpesvirus DNA polymerase inhibitor): the guanine base and acyclic sugar mimic carry the largest attributions, consistent with the nucleoside analogue chain-termination mechanism. Saliency is computed in closed form as $\mathrm{attr}_k(i,j) = [W_m^\top \hat{b}_j^{(t)}]_k$, a single matrix–vector product (linear approximation; see §3.2).

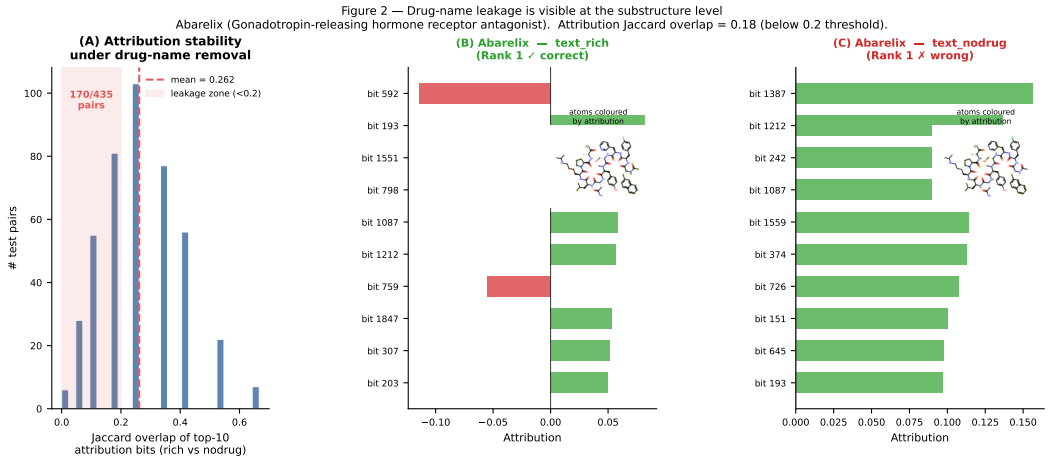

Figure 4: **Drug-name leakage is visible at the substructure level.** **(A)** Distribution of Jaccard overlap between the top-10 attribution bits under `text_rich` and `text_nodrug` across all 435 test pairs. The mean overlap of 0.262 and the 39% of pairs in the leakage zone ($J < 0.2$, shaded) indicate that a large fraction of test-pair attributions are not stable under drug-name removal. **(B, C)** Case study for Abarelix (GnRH receptor antagonist), a leakage candidate with $J = 0.08$. Under `text_rich` (panel B, correct Rank-1 retrieval) the attribution concentrates on the aromatic ring system coupled to the peptide backbone; under `text_nodrug` (panel C, wrong retrieval) attribution redistributes substantially to different structural environments. Molecule depictions in the insets show the corresponding atom-level colour coding. The shift in attribution fingerprint provides structural-level evidence that the model exploits name-correlated molecular features under the full text condition.

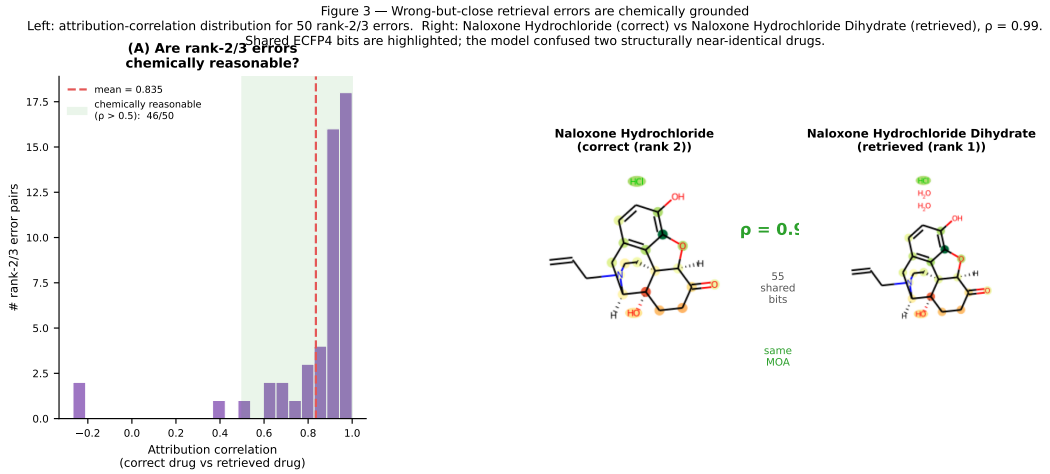

Figure 5: **Wrong-but-close retrieval errors are chemically grounded.** **(A)** Distribution of Pearson attribution correlation ρ_i between the correct drug and the wrongly retrieved drug across 49 rank-2/3 errors (same-drug salt/hydrate pairs excluded; see text). The mean $\rho = 0.609$, with 73% of pairs in the chemically reasonable zone ($\rho > 0.5$, shaded), indicating that the model systematically confuses pharmacologically similar drugs rather than making arbitrary errors. **(B)** Representative example: Dextroamphetamine (correct, rank 2) vs. Amphetamine (retrieved, rank 1), $\rho = 0.945$, same MOA (catecholamine release). Both molecules share the catecholamine pharmacophore ECFP4 bits and differ only in the presence of the dextro stereocenter. The central panel reports the attribution correlation, the number of shared ECFP4 bits, and the matching mechanism of action (MOA). Atoms are coloured by their attribution value on each respective molecule; the near-identical colour patterns confirm that both drugs are attributed identically by the model.

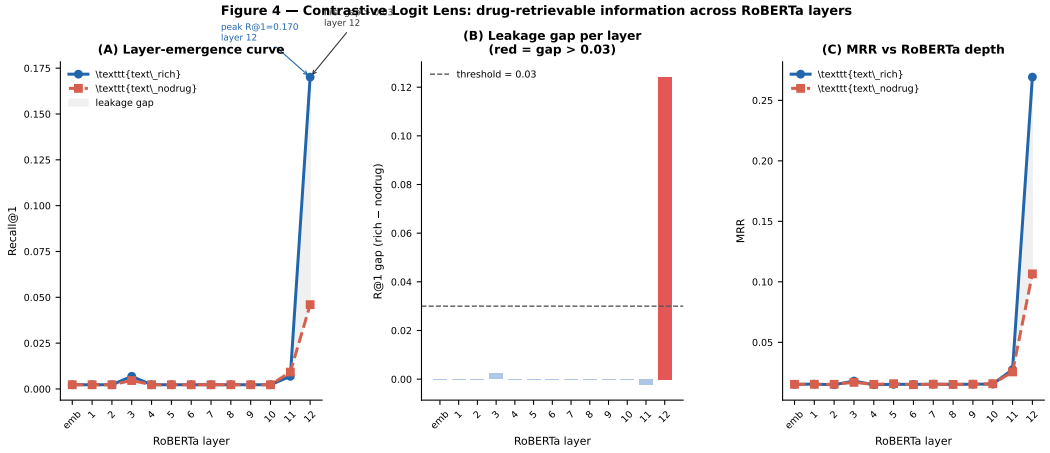

Figure 6: **Contrastive logit lens: drug-retrievable information across RoBERTa layers. (A)** Recall@1 vs. RoBERTa layer for `text_rich` (blue) and `text_nodrug` (red). Both conditions are at chance (R@1 ≈ 0.002) for layers 0–11 and jump sharply at layer 12, where `text_rich` reaches 0.170 and `text_nodrug` reaches 0.046. The grey shading shows the leakage gap; the entire gap of 0.124 is concentrated at the final layer. **(B)** Per-layer leakage gap (R@1 rich − R@1 nodrug): the bar at layer 12 is coloured red (exceeds 0.03 threshold); all other layers are below threshold. **(C)** Distribution of the first layer at which each drug is correctly retrieved, split by whether the drug is in the final correct set (blue) or not (orange). Correct retrievals almost exclusively "lock in" at layer 12 (mean $= 11.9$), while the few intermediate-layer hits for incorrect drugs are noise.

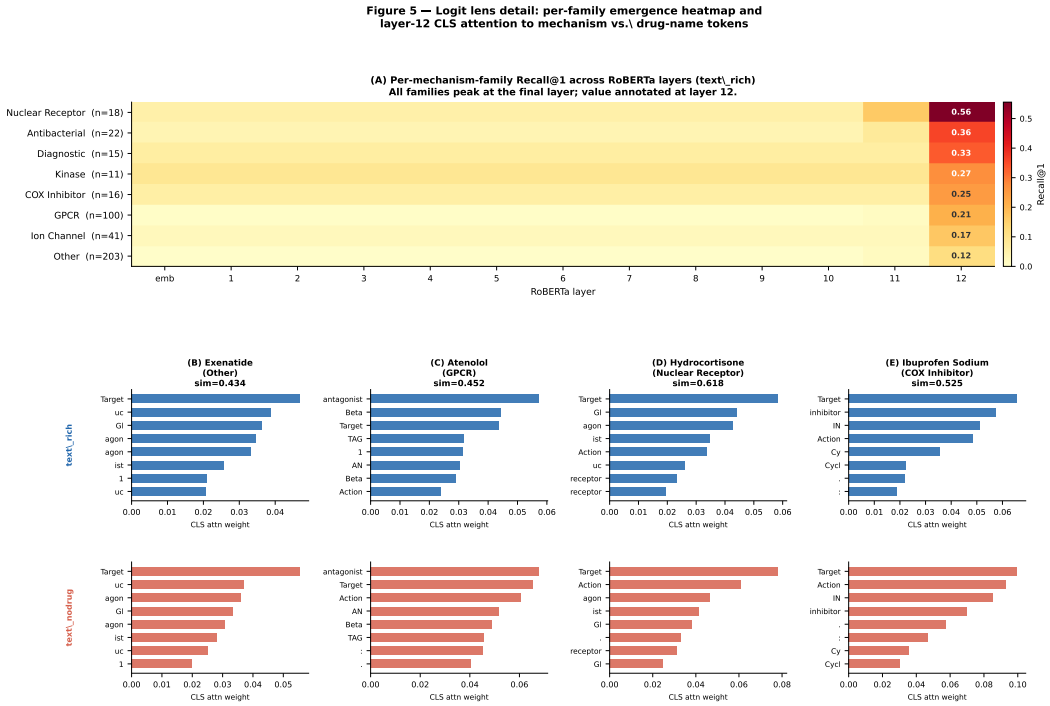

Figure 7: **Logit lens detail: per-family emergence heatmap and layer-12 CLS attention. (A)** Heatmap of Recall@1 at each RoBERTa layer for each mechanism family (`text_rich`). Rows are ordered by descending final-layer performance. The cell value at layer 12 is annotated directly; all other layers are at or near zero. Nuclear receptor ligands achieve the highest retrieval (R@1 $= 0.556$) due to stereotyped mechanism text and structurally distinct steroid scaffolds. GPCRs, the largest family ($n = 129$), reach only 0.209 because GPCR subtypes share similar mechanism text across hundreds of structurally related ligands. **(B)–(E)** CLS token attention weights at layer 12 for four correctly retrieved drugs spanning different families (Exenatide, Atenolol, Hydrocortisone, Ibuprofen sodium). Each drug is shown under both `text_rich` (blue, top row) and `text_nodrug` (orange, bottom row). The top-8 tokens by attention weight are displayed after removing special tokens (`<s>`, `</s>`). Under `text_rich`, attention distributes across punctuation, field delimiters, and drug-name subword tokens. Under `text_nodrug`, drug-name tokens are absent and attention shifts toward mechanism tokens (*AGONIST*, *ANTAGONIST*, *INHIBITOR*) and structural field markers (*Target*, *Action*). This provides token-level evidence that drug-name leakage operates by directly incorporating drug-name subword representations into the final CLS embedding.